

Natural Progression — FOH EQ Rationale

Fountain Square (outdoor) · DiGiCo Quantum 225 · 2026-07-25 · Rev 1.0 Deep Think

Artist. Per Brian's call, processed as R&B (no artist dig). Standard R&B/funk-soul band: full drum kit, DI+cab bass, four guitars, keys, layered vocals, DJ/playback.

Room. Open-air Fountain Square — no room gain, low end + HF dissipate over the throw, façade slap but no reverb tail. Cuts run deeper than indoor (−6 to −9 typ), HPF decisive, clarity first. L-Acoustics A15/KS21 house PA, M32 monitors.

Research. Genre fixed R&B by Brian (no artist research). Per-source fresh web pass (Shure/Audix/Audio-Technica/Sennheiser specs + Gearspace/PSW + masteringthemix/musicguymixing R&B-drum consensus) cross-checked vs KB mic-library + eq-starting-points, and beside Brian's console-verified Izzy Escobar 2.0 build (same FSQ house kit). Backline (Backstage Backline pull 26-0420): Mapex Saturn V kit (22" kick, 10/12/16 toms, 14" Black Panther snare), Sabian HHX cymbals, Yamaha CP33 keys. Weather (Open-Meteo, 6–10pm): 7/25 77–83°F, 43–61% RH, wind 4–8 mph, rain ~1% — HF carries a touch better; don't over-boost the top.

What changed from the KB default / prior rev — and why

- Kick-out D6 (vs Izzy's 52A): near-flat — NO 3–5k boost (capsule bakes +5k) and NO 600 cut (already −15 scooped).
- Added snare-bottom e604: HPF 180, owns wire snap only; i5 keeps body/crack.
- OH is a Beta 27 STEREO pair on fader 9 (not SM81 split 9/10) — top already baked, so no HF boost, tame 9k; ch 10 stays the PL8 return.
- Bass two-mic INVERTED vs Izzy: PG52 (kick mic on the cab) owns sub/thump, DI owns definition — de-stacked the DI lows.
- Four guitars: e609 (1/2) voiced against SM57 (3/4) so the mids don't pile.
- Vocals override the FSQ template's wireless curve/feedback notch — ring out at soundcheck.

Question round — what was asked, what Brian decided

- Q: Percussion 1/2 instrument? -> Brian: generic aux perc — conservative, neutral voicing.
- Q: snare-bottom '604'? -> Brian: Sennheiser e604.
- Genre fixed R&B for all three by Brian; no artist research.

Reverb — Seventh Heaven Pro (100% wet returns)

- **VOCAL — Plates 1 / #06 Vocal Plate** | Settings: Decay 1.5s (factory) · PreDelay 24ms (factory) · VLF −19 (factory) · E/L Max Early · Late −6 (raised from a Memo-deep cut — outdoors carries the late field) · Late Rolloff 6400 (factory) | In-plugin EQ: LC 180 Hz on the return; ducker Reverb mode, Thr −22 dB | Why: Lead-vocal sheen + presence, sits forward over the band; no room outdoors to muddy it — the workhorse lead plate.
- **VOCAL — Chambers 1 / #10 Vocal Chamber** | Settings: Decay 1.2s (from 1.6s factory — tighter for up-tempo funk) · PreDelay 15ms (from 0 factory) · VLF −15 (from −10 factory — less low wash outdoors) · E/L Max Early · Late −8 · Late Rolloff 8000 (factory) | In-plugin EQ: LC 200 Hz on the return | Why: Gives place without wash on up-tempo/funk material; hands off to the Vocal Plate when a song slows down.
- **VOCAL — Halls 1 / #11 Gold Hall** | Settings: Decay 1.8s (factory) · PreDelay 20ms · VLF −12 (cut hard from 0 factory — Gold Hall always gets its VLF cut) · E/L Max Early · Late −6 · Late Rolloff 6000 (from factory) | In-plugin EQ: LC 200 Hz + HS −2 above 8k on the return | Why: The ballad / feature bloom when a song opens up — used sparingly, the big-moment size.

- **INSTRUMENT — Plates 1 / #04 Snare Plate A** | Settings: Decay 1.0s (from 1.2s factory — tighter outdoors) · PreDelay 0 (factory) · VLF -15 · E/L Max Early · Late -8 · Late Rolloff 6400 (factory) | In-plugin EQ: LC 250 Hz on the return | Why: Adds snap/depth to the snare on the funk backbeat without washing the kit. Feeds the ch-10 PL8 return.
- **GENERAL — Rooms 1 / #02 Studio B Close** | Settings: Decay 0.6s (factory) · PreDelay 0 · VLF -12 (cut hard from 0 factory — always cut) · E/L Max Early · Late -10 · Late Rolloff 6800 (factory) | In-plugin EQ: LC 300 Hz on the return | Why: A hair of glue on the kit/percussion so the outdoor stage isn't bone-dry — very short, no obvious tail.
- **USING THEM TOGETHER** — Vocal Plate is the default lead — forward and present. On up-tempo/funk numbers switch the lead to the tighter Vocal Chamber so the groove stays dry and articulate; the two hand off song-to-song. Gold Hall is the big-moment throw only — a feature or a ballad — never under the whole set. Snare Plate A feeds the reserved ch-10 PL8 return for backbeat depth; Studio B Close is a whisper of kit glue only. Everything runs 100% wet on its own return, VLF cut, Late higher than a Memo send because there's no room to carry the tail.

DRUMS

Ch 1 | Kick In · Shure Beta 91A

HPF 40 | LPF off || B4 +4@3 kHz Q1.2 BELL B3 -3@400 Hz Q2 BELL B2 -3@250 Hz Q1.8 BELL B1 +2@80 Hz Q1.2 BELL

Beta 91A boundary inside the kick — the click/attack engine. Contour assumed FLAT (the 400 cut lives on the desk; if the switch is engaged, halve it). Owns click + upper; lows tucked under the D6.

+4@3k beater definition that travels outdoors (R&B punch, not a metal click). 400/250 clear the shell boom the boundary picks up — eased for open air. Lows modest (+2@80) so they don't stack with the D6's baked 63Hz. TRACE: base(91A boundary, +4@3k click) · equip(22" Mapex Saturn kick — no change) · genre(R&B — punch over click) · artist(no dig) · venue(FSQ — box cuts eased, no room mud). COMP (doc only, not patched): Green FET ~4:1, 15ms atk, 3–4 dB GR.

Ch 2 | Kick Out · Audix D6

HPF 40 | LPF off || B4 flat B3 flat B2 -2@250 Hz Q1.8 BELL B1 +3@60 Hz Q1.2 BELL

Audix D6 pre-scooped 'smiley' — peaks ~63Hz & ~5kHz, deep scoop -15dB @600. Thump+click baked in; needs almost no EQ. Owns the sub/body under the 91A's click.

Barely touched — the capsule IS the kick voicing. +3@60 supports the low open air swallows; light 250 trim. Capsule-voicing gate: NO 3–5k boost (D6 already bakes +5k — a boost stacks harsh) and NO 600 cut (already -15 scooped). TRACE: base(D6 smiley — near-flat) · equip(22" kick — no change) · genre(R&B thump) · artist(—) · venue(FSQ — +60 for lost low). Two-mic: D6 owns sub, 91A owns click.

Ch 3 | Snare Top · Audix i5

HPF 120 | LPF off || B4 +2@7 kHz Q0.8 BELL B3 -2@5.5 kHz Q2 BELL B2 +3@1 kHz Q1 BELL B1 -5@300 Hz Q2 BELL

i5 bakes +5dB@150 body + a strong +9dB@5.5k presence, mids slightly scooped — more open/body than a 57. Owns the snare body + crack; the e604 owns the wire snap.

Capsule-voicing gate: the +9@5.5k is baked, so I TRIM it -2 (not boost) and put the snap up at 7k where the capsule rolls off. +3@1k crack, -5@300 box (eased — the i5 isn't boxy, don't gut it). TRACE: base(i5 +9@5.5k baked → trim) · equip(14" Black Panther — no change) · genre(R&B backbeat — body over crack) · artist(—) · venue(FSQ — 7k snap to cut outdoors). GATE (doc): fast ~1ms atk but ghost notes/press rolls must live — 15–20ms rel. Two-mic: i5 body+crack, e604 snap.

Ch 4 | Snare Bot · Sennheiser e604

HPF 180 | LPF off || B4 +3@8 kHz Q0.8 BELL B3 -3@1 kHz Q1.5 BELL B2 flat B1 flat

e604 clip on the bottom head — scooped low-mids, voiced attack, thin lows (ideal for wires). Owns the snare-wire snap/sizzle only.

HPF way up at 180 so it's wires, not shell or kick bleed. +3@8k snap; -3@1k clears the e604's box. Lane: owns the top-end wire sizzle, the i5 owns body — no overlap. TRACE: base(e604 thin/voiced-attack) · equip(snare wires) · genre(R&B — crisp backbeat) · artist(—) · venue(FSQ — 8k to cut). GATE (doc): key off the snare top so wires don't chatter.

Ch 5 | Hat · Shure SM81

HPF 500 | LPF 16000 || B4 flat B3 flat B2 -5@2 kHz Q1.2 BELL B1 -6@800 Hz Q1.5 BELL

SM81 — ruler-flat, un-hyped highs. Takes EQ straight; no built-in flattery to fight.

No HF boost (81 is already detailed and dry 7/24 air thins the top over the throw anyway — don't make it brittle). Clear the low bleed (800) + the 2k clang so it sparkles clean. TRACE: base(SM81 flat) · equip(14" HHX hats) · genre(R&B — tight hat) · artist(—) · venue(FSQ+dry — resist top boost).

Ch 6 | Rack 1 · Audix D2

HPF 90 | LPF 12000 || B4 flat B3 +4@2.8 kHz Q1 BELL B2 -3@450 Hz Q1.8 BELL B1 +3@135 Hz Q1.2 BELL

D2 tuned to control tom resonance — articulate with little EQ; leaner deep lows than the D4.

Attack lift +4@2.8k (trust the capsule), gentle 135 body, modest 450 mud cut. TRACE: base(D2 — articulate) · equip(10" Saturn rack) · genre(R&B fills) · artist(—) · venue(FSQ). GATE (doc): snappy release for funk; don't choke the decay.

Ch 7 | Rack 2 · Audix D2

HPF 80 | LPF 12000 || B4 flat B3 +4@2.5 kHz Q1 BELL B2 -3@405 Hz Q1.8 BELL B1 +3@120 Hz Q1.2 BELL

As Rack 1, shifted down for the larger 12" drum.

Let the D2 do the work; body at 120 for the bigger shell. TRACE: base(D2) · equip(12" Saturn rack) · genre · artist(—) · venue(FSQ).

Ch 8 | Floor · Audix D4

HPF 55 | LPF 10000 || B4 flat B3 +4@1.8 kHz Q1 BELL B2 -3@405 Hz Q1.8 BELL B1 +4@70 Hz Q0.71 SHELF

D4 reaches ~an octave lower than the D2 natively; smooth 800–1k (not scooped) — may want a touch of click.

Body shelf modest +4@70 (support, not a lift; ride the sub outdoors). +4@1.8k stick definition the D4 lacks. TRACE: base(D4 — deep, smooth mids) · equip(16" Saturn floor) · genre(R&B) · artist(—) · venue(FSQ — modest sub). GATE (doc): slower release than the racks so the floor decay lives.

Ch 9 | OH (st) · Shure Beta 27

HPF 200 | LPF off || B4 flat B3 flat B2 -2@9 kHz Q2 BELL B1 -5@300 Hz Q1.5 BELL

Beta 27 LDC pair as overheads — flat 60Hz–3k with baked +2dB bumps at 5.5k & 9k, supercardioid (tight, less room — good outdoors), –15 pad available.

Capsule-voicing gate: the top is already lifted (5.5k+9k baked) → NO HF boost; a light –2@9k tames fizz instead. –5@300 clears low wash + kit bleed so it's cymbals/sheen, not a room mic. Supercardioid keeps the outdoor stage tight. HPF 200. TRACE: base(Beta 27 — baked top) · equip(HHX cymbals) · genre(R&B — cymbal sheen) · artist(—) · venue(FSQ — supercard tight, clear wash).

BASS

Ch 11 | Bass DI · XLR (DI)

HPF 40 | LPF off || B4 -4@1.5 kHz Q1.5 BELL B3 +4@700 Hz Q1 BELL B2 -5@250 Hz Q2 BELL B1 +3@80 Hz Q0.71 SHELF

Bass DI — clean, full-range. The PG52 (a kick mic) is on the cab. LANE: the DI owns note definition + mids + upper harmonics; the PG52 owns the sub/thump — so the DI's lows are pulled back to de-stack.

R&B/soul bass = round, low-mid forward, top rolled. +4@700 note/growl (DI's lane), –4@1.5k tames string clank, –5@250 de-stacks off the PG52's low body, +3@80 modest (PG52 carries the real sub). TRACE: base(DI clean — definition lane) · equip(bass rig, DI+cab) · genre(R&B — round, low-mid forward) · artist(—) · venue(FSQ — support 80). COMP (doc): opto ~3:1, 20–30ms atk, even the pocket.

Ch 12 | Bass Mic · Shure PG52

HPF 30 | LPF 8000 || B4 flat B3 +2@2.5 kHz Q1 BELL B2 -5@400 Hz Q2 BELL B1 +4@90 Hz Q0.71 SHELF

PG52 is a kick mic on the bass cab — 30Hz–13k, tuned low-end thump + a slight upper-mid click bump, rolls off above 13k (no air). It's the SUB/BODY layer under the DI.

Owens the low thump: +4@90 shelf. -5@400 carves a pocket so it doesn't double the DI's mud. Small +2@2.5k for finger attack (the PG52's click bump). LPF 8k — no useful top, keep it dark body. TRACE: base(PG52 kick-mic-on-cab — sub lane) · equip(bass cab) · genre(R&B thump) · artist(—) · venue(FSQ). Two-mic: PG52 lows, DI definition — no low boost on both.

GUITAR

Ch 13 | Gtr 1 · Sennheiser e609

HPF 100 | LPF off || B4 -3@3 kHz Q2 BELL B3 +2@1.2 kHz Q1.2 BELL B2 -4@400 Hz Q2 BELL B1 -3@150 Hz Q1.5 BELL

e609 flat-face over the cab — bright upper-mid/presence 4–5k, supercardioid, can get edgy 2.5–4k on bright amps. Owns cab presence.

Rhythm R&B guitar sits under the vocal + horns. -3@3k tames the e609 edge; +2@1.2k so the comp speaks; -4@400 box; HPF 100 + -3@150 keeps it out of the bass. TRACE: base(e609 — bright/edgy) · equip(gtr cab) · genre(R&B comp — supportive, not a wall) · artist(—) · venue(FSQ — deeper box cut). Staggered vs the 57 guitars.

Ch 14 | Gtr 2 · Sennheiser e609

HPF 100 | LPF off || B4 -3@2.8 kHz Q2 BELL B3 +2@1.4 kHz Q1.2 BELL B2 -4@450 Hz Q2 BELL B1 -3@150 Hz Q1.5 BELL

e609 as Gtr 1; voiced a hair different (edge 2.8k, presence 1.4k) so two e609 guitars don't overlay.

As Gtr 1 with the edge/presence shifted so the two e609 guitars separate. TRACE: base(e609) · equip(gtr cab) · genre(R&B comp) · artist(—) · venue(FSQ).

Ch 15 | Gtr 3 · Shure SM57

HPF 100 | LPF off || B4 +3@3.5 kHz Q1 BELL B3 -2@800 Hz Q1.2 BELL B2 -5@400 Hz Q2 BELL B1 -3@180 Hz Q1.5 BELL

SM57 on the cab — mid-forward, presence 3–5k, builds box/honk 300–500. Owns mid grit.

+3@3.5k presence (57's lane — differs from the e609 guitars' voicing so 4 gtrs separate), -5@400 box, -2@800 honk, HPF 100 + 180 trim off the bass. TRACE: base(SM57 — mid grit) · equip(gtr cab) · genre(R&B — present, under the vocals) · artist(—) · venue(FSQ — deeper box).

Ch 16 | Gtr 4 · Shure SM57

HPF 100 | LPF off || B4 +3@3 kHz Q1 BELL B3 -2@850 Hz Q1.2 BELL B2 -5@450 Hz Q2 BELL B1 -3@180 Hz Q1.5 BELL

SM57 as Gtr 3; presence at 3k, box 450 so the two 57 guitars separate.

As Gtr 3, shifted. TRACE: base(SM57) · equip(gtr cab) · genre(R&B) · artist(—) · venue(FSQ).

KEYS

Ch 17 | Keys 1 · XLR (DI)

HPF 50 | LPF off || B4 flat B3 +3@2.5 kHz Q0.8 BELL B2 +2@300 Hz Q0.8 BELL B1 -5@200 Hz Q1.5 BELL

Yamaha CP33 stage piano via DI, stereo pair (EP/organ/piano patches).

Warmth +2@300 + clarity +3@2.5k without hype; -5@200 clears mud off the bass/vocal low-mids. Stereo — keep the pair identical. TRACE: base(CP33 DI) · equip(CP33 88-key) · genre(R&B keys) · artist(—) · venue(FSQ).

Ch 18 | Keys 2 · XLR (DI)

HPF 50 | LPF off || B4 flat B3 +3@2.5 kHz Q0.8 BELL B2 +2@300 Hz Q0.8 BELL B1 -5@200 Hz Q1.5 BELL

Yamaha CP33 stage piano via DI, stereo pair (EP/organ/piano patches).

Warmth +2@300 + clarity +3@2.5k without hype; -5@200 clears mud off the bass/vocal low-mids. Stereo — keep the pair identical. TRACE: base(CP33 DI) · equip(CP33 88-key) · genre(R&B keys) · artist(—) · venue(FSQ).

DRUMS

Ch 19 | Drum Pad · XLR (DI)

HPF 30 | LPF off || B4 flat B3 flat B2 flat B1 flat

Electronic drum pad (SPD-style) via XLR — pre-voiced samples (claps/subs/percussion).

Samples are pre-produced — near-flat, HPF 30 to protect the sub hits. Trim live only if a sample clouds the low-mids. TRACE: base(pre-voiced samples) · venue(FSQ).

AMBIENT

Ch 20 | Stem 1 · XLR (DI)

HPF 30 | LPF off || B4 flat B3 flat B2 flat B1 flat

Backing-track stem 1 — mastered, full-range.

Already balanced — EQ like a mic thins the production. Near-flat, HPF 30. Small 315 trim live only if it clouds the band. TRACE: base(mastered stem) · venue(FSQ).

Ch 21 | Stem 2 · XLR (DI)

HPF 30 | LPF off || B4 flat B3 flat B2 flat B1 flat

Backing-track stem 2 — mastered, full-range.

Already balanced — EQ like a mic thins the production. Near-flat, HPF 30. Small 315 trim live only if it clouds the band. TRACE: base(mastered stem) · venue(FSQ).

Ch 22 | Stem 3 · XLR (DI)

HPF 30 | LPF off || B4 flat B3 flat B2 flat B1 flat

Backing-track stem 3 — mastered, full-range.

Already balanced — EQ like a mic thins the production. Near-flat, HPF 30. Small 315 trim live only if it clouds the band. TRACE: base(mastered stem) · venue(FSQ).

Ch 23 | Stem 4 · XLR (DI)

HPF 30 | LPF off || B4 flat B3 flat B2 flat B1 flat

Backing-track stem 4 — mastered, full-range.

Already balanced — EQ like a mic thins the production. Near-flat, HPF 30. Small 315 trim live only if it clouds the band. TRACE: base(mastered stem) · venue(FSQ).

Ch 24 | BT Aux · Line / BT

HPF 40 | LPF off || B4 flat B3 flat B2 flat B1 flat

Bluetooth aux — walk-in / between-set music, not a performance input.

Near-flat playback. HPF 40 only. TRACE: base(mastered playback) · venue(FSQ).

VOCALS

Ch 25 | Wls 1 · Shure Beta 58A

HPF 100 | LPF 16000 || B4 -3@10 kHz Q2 BELL +DEQ B3 -2@4 kHz Q2.5 BELL B2 -4@600 Hz Q1.5 BELL B1 flat

Beta 58A supercardioid handheld — TWO baked presence peaks (~4k & ~10k), boxy ~600, <500 attenuated, tight pattern/high GBF (brighter than an SM58). LEAD voicing.

Cut-only (feedback + taste). Dynamic de-ess -3@10k on the baked top peak so the air stays between sibilants; light -2@4k tames stridency without killing the bite the capsule bakes; -4@600 box for outdoor intelligibility. HPF 100. Overrides the FSQ template wireless curve/notch — ring out at soundcheck; reach for comp + a touch of plate before more EQ. TRACE: base(Beta 58A twin peaks) · genre(R&B lead) · venue(FSQ intelligibility).

Ch 26 | Wls 2 · Shure Beta 58A

HPF 120 | LPF 16000 || B4 -2@10 kHz Q2 BELL +DEQ B3 -2@3 kHz Q2.5 BELL B2 -4@600 Hz Q1.5 BELL B1 flat

Beta 58A handheld — backing voicing: sits a touch behind the lead (-2@3k) so the harmony supports.

Cut-only. -4@600 box, -2@10k de-ess, -2@3k to seat it behind the lead. HPF 120. Overrides the template wireless notch — ring out. TRACE: base(Beta 58A) · genre(R&B BGV) · venue(FSQ).

Ch 27 | Wls 3 · Shure Beta 58A

HPF 120 | LPF 16000 || B4 -2@10 kHz Q2 BELL +DEQ B3 -2@3 kHz Q2.5 BELL B2 -4@600 Hz Q1.5 BELL B1 flat

Beta 58A handheld — backing voicing: sits a touch behind the lead (-2@3k) so the harmony supports.

Cut-only. -4@600 box, -2@10k de-ess, -2@3k to seat it behind the lead. HPF 120. Overrides the template wireless notch — ring out. TRACE: base(Beta 58A) · genre(R&B BGV) · venue(FSQ).

Ch 28 | Wls 4 · Shure Beta 58A

HPF 120 | LPF 16000 || B4 -2@10 kHz Q2 BELL +DEQ B3 -2@3 kHz Q2.5 BELL B2 -4@600 Hz Q1.5 BELL B1 flat

Beta 58A handheld — backing voicing: sits a touch behind the lead (-2@3k) so the harmony supports.

Cut-only. -4@600 box, -2@10k de-ess, -2@3k to seat it behind the lead. HPF 120. Overrides the template wireless notch — ring out. TRACE: base(Beta 58A) · genre(R&B BGV) · venue(FSQ).

Ch 29 | Vox 5 · Shure SM58

HPF 110 | LPF 16000 || B4 flat B3 -2@5 kHz Q2.5 BELL B2 -4@500 Hz Q1.5 BELL B1 flat

SM58 on a stand — presence 4–6k, boxy ~500, gentle low rolloff. Backing/utility vocal.

Cut-only. -4@500 box, light -2@5k if spitty, HPF 110. Supportive — a touch behind the leads. TRACE: base(SM58) · genre(R&B BGV) · venue(FSQ).

Ch 30 | Vox 6 · Shure SM58

HPF 110 | LPF 16000 || B4 flat B3 -2@5 kHz Q2.5 BELL B2 -4@500 Hz Q1.5 BELL B1 flat

SM58 on a stand — presence 4–6k, boxy ~500, gentle low rolloff. Backing/utility vocal.

Cut-only. -4@500 box, light -2@5k if spitty, HPF 110. Supportive — a touch behind the leads. TRACE: base(SM58) · genre(R&B BGV) · venue(FSQ).

AMBIENT

Ch 31 | DJ L · XLR (DI)

HPF 30 | LPF off || B4 flat B3 flat B2 flat B1 flat

DJ playback (turntables/controller), stereo — mastered full-range.

Left near-flat — EQ'ing mastered playback like a mic thins it. HPF 30 only; add a small 315 trim live only if it clouds the band. TRACE: base(mastered stereo) · venue(FSQ).

Ch 32 | DJ R · XLR (DI)

HPF 30 | LPF off || B4 flat B3 flat B2 flat B1 flat

DJ playback (turntables/controller), stereo — mastered full-range.

Left near-flat — EQ'ing mastered playback like a mic thins it. HPF 30 only; add a small 315 trim live only if it clouds the band. TRACE: base(mastered stereo) · venue(FSQ).

Deep-research pass — values reasoned from mic behavior, instrument, genre, the artist's references, and the room. Informed starting points, not gospel; trust your ears at soundcheck.